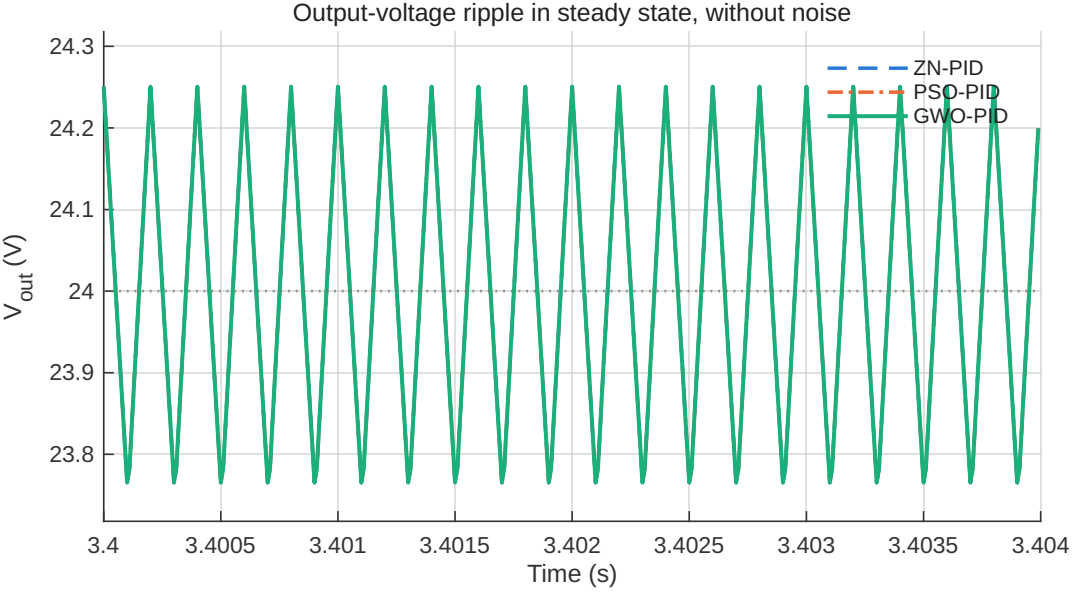
Output-voltage ripple in steady state, without noise
ZN-PID
PSO-PID
GWO-PID
24.3
24.2
24.1
24
23.9
23.8
V out (V)
3.4
3.4005
3.401
3.4015
3.402
3.4025
3.403
3.4035
3.404
Time (s)